

Industrial IoT SaaS Platform

Engineering Handbook - Volume 01

Vision & Product Requirements

Version 1.0

1. Document Purpose

This handbook is intended to become the single source of truth for the Energy Management Industrial IoT SaaS Platform. It should allow a new engineer or AI coding assistant to understand the project without relying on previous chat history.

2. Problem Statement

Factories often operate with disconnected systems. Energy, production, water, maintenance, recipes and costing are isolated. Managers lack a unified platform for monitoring assets, analysing telemetry and making decisions. The platform aims to solve this through a configurable Industrial IoT SaaS architecture.

3. Long-Term Vision

Build a cloud-native Industrial IoT platform capable of serving thousands of factories while remaining industry agnostic. Textile processing is the first domain, but the architecture must support food, chemicals, pharmaceuticals and manufacturing.

4. Business Goals

Provide real-time monitoring, historical analytics, alarms, predictive maintenance, production intelligence, utility management and mobile access. Minimise manual data entry and support AI-assisted operations.

5. Guiding Principles

1. Multi-tenant from day one.
2. Configuration over hardcoding.
3. API-first architecture.
4. Docker-first deployment.
5. Security by default.
6. Documentation evolves with code.
7. Every design decision should be explainable.

6. Primary Users

Factory Owners
Plant Managers
Maintenance Engineers
Production Supervisors
Energy Managers

Machine Operators
System Administrators

7. Current Scope

Completed foundation includes Docker deployment, FastAPI backend, PostgreSQL database, authentication, company/user hierarchy, machine hierarchy, initial telemetry models, GitHub repository and project documentation.

8. Future Scope

Telemetry ingestion, MQTT gateway, Raspberry Pi collector, dashboards, alarms, predictive maintenance, OCR recipe extraction, costing engine, reporting, Flutter mobile applications and SaaS billing.

9. Technology Choices

FastAPI was selected for high performance and clean API design. PostgreSQL provides relational integrity. Docker simplifies deployment. GitHub provides version control. Raspberry Pi acts as the edge gateway.

10. AI Collaboration Strategy

Every future AI assistant must read the documentation before modifying the codebase. AI should preserve backward compatibility, document changes, explain code, and update the handbook after every major architectural change.

Roadmap Snapshot

- Phase 1 - Platform Foundation
- Phase 2 - Asset Configuration
- Phase 3 - Telemetry Ingestion
- Phase 4 - Dashboards & Analytics
- Phase 5 - Alarm Engine
- Phase 6 - Mobile Applications
- Phase 7 - Predictive Maintenance
- Phase 8 - Textile OCR & Costing
- Phase 9 - Commercial SaaS Platform

This is Volume 01 of the Engineering Handbook. Subsequent volumes will cover System Architecture, Database Design, API Specifications, Telemetry, MQTT, Device Layer, Deployment, Security and AI Development Guidelines in detail.